

MATH 3115 OMS & OED COMPLEX ANALYSIS

Lecture 6: The Cauchy Goursat Theorem

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Lesson Outcome

By the end of this lesson you should be able to:

- Define Integral Triangle Inequality .
- Apply Integral Triangle Inequality in evaluating contour integrals.
- Describe Simply connected Domain
- Describe multiply connected Domain
- Solve problems involving cauchy Goursat theorem

Integral Triangle Inequality

Lemma 1

If $f(t) = u(t) + iv(t)$ is a continuous function of a real parameter t , then

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt$$

Integral Triangle Inequality CONT'D

Proof.

Let $f(t)=u(t)+i v(t)$, where $u(t)$ and $v(t)$ are real-valued functions of the real parameter t , and i is an imaginary unit.

If we

Recall the modulus of the complex function $f(t)$:

$$|f(t)| = \sqrt{u(t)^2 + v(t)^2}$$

Now, we wish to prove that:

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt$$

Integral Triangle Inequality CONT'D

Proof.

The integral of $f(t) = u(t) + iv(t)$ over $[a, b]$ can be written as:

$$\int_a^b f(t)dt = \int_a^b u(t)dt + i \int_a^b v(t)dt. \text{ (decomposition)}$$

Let real and imaginary parts of the integral be denoted as follows:

$$I_1 = \int_a^b u(t)dt \quad \text{and} \quad I_2 = \int_a^b v(t)dt$$

Proof.

Then, we have:

$$\int_a^b f(t)dt = I_1 + iI_2$$

Since the modulus of a complex number $z = I_1 + iI_2$ is given by:

$$|I_1 + iI_2| = \sqrt{I_1^2 + I_2^2}$$

By the triangle inequality for integrals:

$$\left| \int_a^b u(t)dt \right| \leq \int_a^b |u(t)|dt \quad \text{and} \quad \left| \int_a^b v(t)dt \right| \leq \int_a^b |v(t)|dt. \text{ Therefore:}$$

$$\sqrt{I_1^2 + I_2^2} \leq \sqrt{\left(\int_a^b |u(t)|dt \right)^2 + \left(\int_a^b |v(t)|dt \right)^2}$$

But from above modulus equation

$$|f(t)| = \sqrt{u(t)^2 + v(t)^2}$$

PROOF CONT'D

Proof.

$$\int_a^b |f(t)| dt \geq \int_a^b \sqrt{u(t)^2 + v(t)^2} dt. \text{ By } \triangle \text{ inequality}$$

Combining all the inequalities, we arrive at:

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt$$



Reading Assignment

Prove Lemma 1 using the value of integral in polar form

ML INEQUALITY OF CONTOUR INTEGRAL

Lemma 2

If $f(z) = u(x, y) + iv(x, y)$ is a continuous function on the contour c , then $\left| \int_c f(z) dz \right| \leq ML$ where L is the length of the contour C and M is upper bound for the modulus $|f(z)|$ on the contour C .

PROOF OF LEMMA 2

Proof.

Using ML inequality and definition of the contour integral, we get the following

$$\left| \int_c f(z) dz \right| \leq \left| \int_a^b f(z(t)) z'(t) dt \right| \leq \int_a^b |f(z(t)) z'(t)| dt.$$

Let M be a positive real constant such that $|f(z)| \leq M$ for all z on c .

Then we have:

$$\left| \int_c f(z) dz \right| \leq \int_a^b M |z'(t)| dt = ML.$$

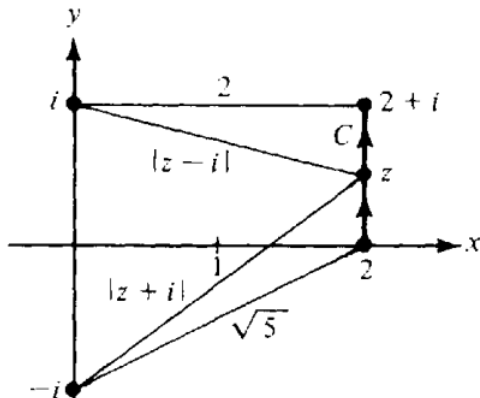
$$\text{Hence } \left| \int_c f(z) dz \right| \leq ML$$



Example 1

Use the ML inequality to show that $\left| \int_C \frac{dz}{z^2+1} \right| \leq \frac{1}{2\sqrt{5}}$
where C is the straight line segment from 2 to $2+i$

solution to example 1



Solution to Example 1 CONT'D

- Here $|z^2 + 1| = |z - i| \times |z + i|$,
- and the terms $|z - i|$ and $|z + i|$ represent the distance from the point z to the points i and $-i$, respectively.
- From the figure above slide and using a geometric argument we see that

$$|z - i| \geq 2 \text{ and } |z + i| \geq \sqrt{5} \text{ for } z \text{ on } C.$$

- Thus: The distance from $-i$ to 2 on x is $\sqrt{2^2 + 1^2} = \sqrt{5}$ (by Pythagorus Theorm or distance formula). So $M = \sqrt{5}$
- L is the length from 2 to $2 + i = |2 + i - 2| = |i| = 1$. Here we have

$$|f(z)| = \frac{1}{|z - i||z + i|} \leq \frac{1}{2\sqrt{5}} = M,$$

and $L = 1$, so ML inequality implies that

$$\left| \int_C \frac{1}{z^2 + 1} dz \right| \leq ML = \frac{1}{2\sqrt{5}}.$$

THE CAUCHY-GOURSAT THEOREM

Theorem 1

The Cauchy-Goursat theorem states that within certain domains the integral of an analytic function over a simple closed contour is zero.

- We will use the technique of partial fractions together with Cauchy-Goursat Theorem to evaluate certain integral.
- Consider the closed contours given below

THE CAUCHY-GOURSAT THEOREM

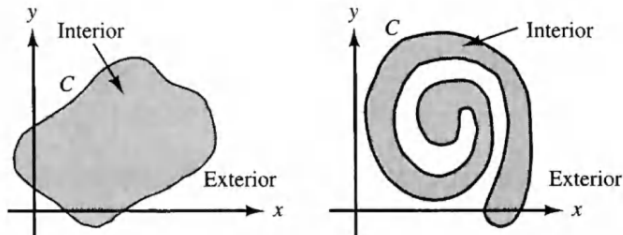


FIGURE The interior and exterior of simple closed contours.

- A domain which is bounded is called the interior of C
- The domain which is unbounded is called the exterior of C

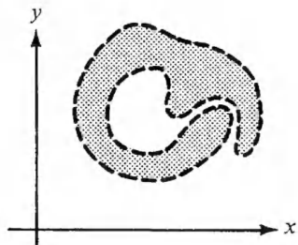
Simply connected Domain

Definition 3

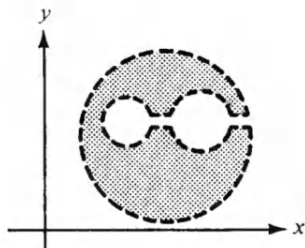
A domain D is said to be simply connected if it has the property that any simple closed contour C contained in D has interior contained in D .

- A domain that has no holes is a simply connected domain

Simply connected Domain CONT'D



(a) A simply connected domain.



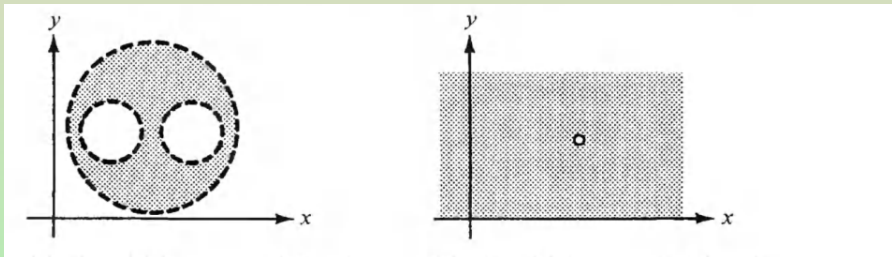
(b) A simply connected domain.

Multiply connected Domain

Definition 4

A domain is said to be multiply connected domain if it is not simply connected

- A domain that has holes is a multiply connected domain



THE CAUCHY-GOURSAT THEOREM 2

Theorem 5

Let f be analytic in a simply connected domain D . If C is a simple closed contour that lies in D , then

$$\int_C f(z) dz = 0$$

Proof.

The proof is left for the reader as an exercise.
Also read about Greens theorem



THE CAUCHY-GOURSAT THEOREM

Example 2

Consider functions given by the following
 $f(z) = e^z$, $g(z) = \cos z$, $h(x) = z^n$. Show that the value of each function is zero using Cauchy-Goursat Theorem.

THE CAUCHY-GOURSAT THEOREM

Solution to Example 2

$f(z) = e^z$, $g(z) = \cos z$, and $h(z) = z^n$ are all entire functions and they have continuous derivatives.

The Cauchy-Goursat Theorem implies that for any simple closed contour, we have

$$\int_C e^z dz = 0, \quad \int_C \cos z dz = 0, \quad \int_C z^n dz = 0$$

THE CAUCHY-GOURSAT THEOREM

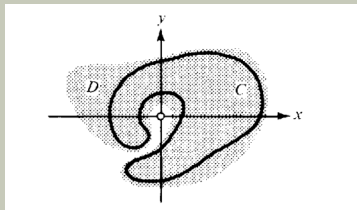
Example 3

If C is a simple closed contour such that the origin does not lie in the interior of C . Show that if $f(z) = \frac{1}{z^n}$, then $\int_C f(z)dz = 0$.

THE CAUCHY-GOURSAT THEOREM

Solution to Example 3

- Consider the diagram below



- If C is a simple closed contour, such that the origin does not lie in the interior of C , then there is a simply connected domain D that contains C in which $f(z) = \frac{1}{z^n}$ is analytic as shown in the figure above. Hence the Cauchy-Goursat Theorem implies that $\int_C \frac{1}{z^n} dz = 0$ since the origin does not lie in C .

THE END